

Spread Locator: A Statistical Distribution Analysis Model

Part A - Theoretical Foundation :-

1. What is Statistical Distributions?

A **statistical distribution** shows:

- how data values are spread
- how often each value occurs
- probability of each outcome

 **Example:**

Marks of students:

[40, 50, 50, 60, 70, 70, 70, 80]

Here:

- 70 appears most → high probability
- 40 appears least → low probability

 **Types:**

- Discrete distribution (countable values)
- Continuous distribution (any value in range)

Real-Life Examples

- 🎯 Exam marks distribution in class
- 📊 Salary distribution in company
- 🚗 Speed of cars on road
- ☎️ Number of calls per hour

2. What is a Q-Q Plot?

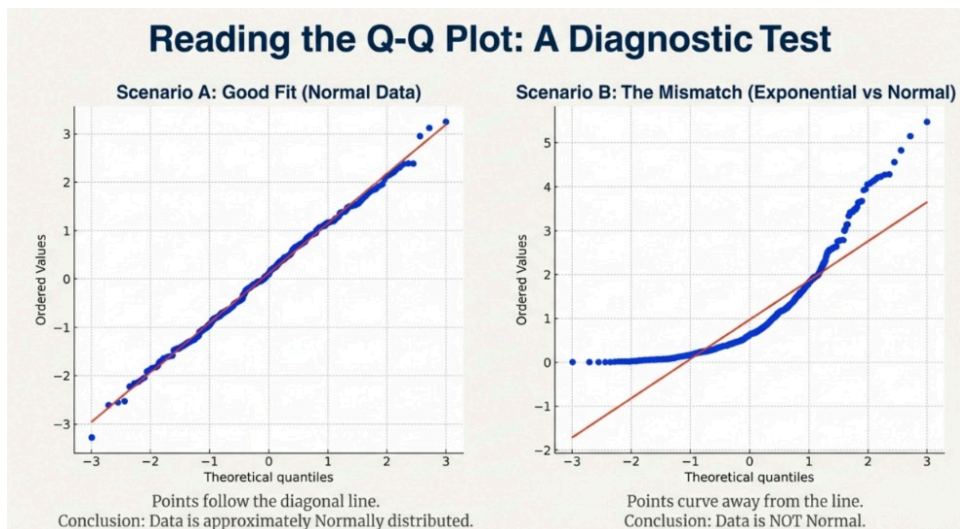
A **Q-Q plot** checks if data follows a theoretical distribution (like normal).

📌 **Use:**

- Check normality of data
- Compare two distributions

📊 **Idea:**

- If points lie on a straight line → data is normal



🐍 Python Code (Q-Q Plot)

Python

```
import numpy as np
import scipy.stats as stats
import matplotlib.pyplot as plt
```

3. Discrete vs Continuous Distribution

Feature	Discrete Distribution	Continuous Distribution
Meaning	Countable values	Measurable values
Values	Whole numbers	Whole numbers and decimals
Nature	Separate values	Any value in a range
Example	Dice, Coins	Height, Weight
Data Type	Counting	Measuring
Graph	Bar Graph	Smooth Curve

4. Bernoulli Distribution

Bernoulli Distribution

 **Definition:**

Only **2 outcomes**:

- Success (1)
- Failure (0)

 **Example:**

- Toss coin $\rightarrow$ Head (1), Tail (0)
- Exam pass/fail

 **Formula:**

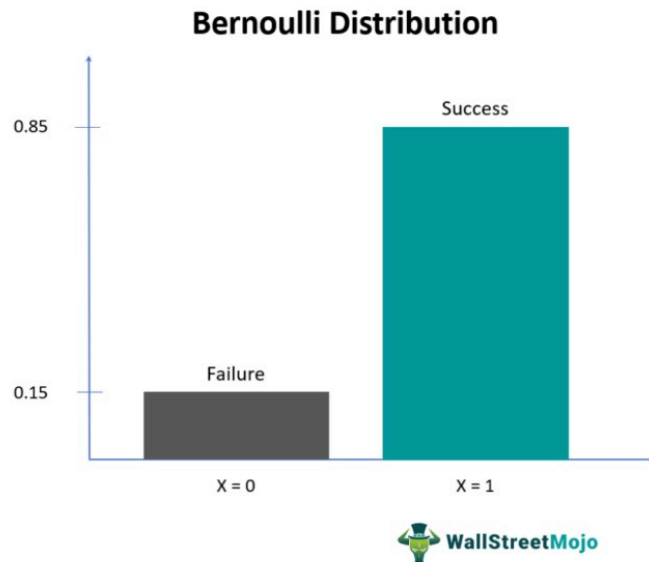
BERNOULLI DISTRIBUTION

- A Bernoulli trial is an experiment with only two outcomes. An r.v. X has Bernoulli(p) distribution if

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}; 0 \leq p \leq 1$$

$$P(X = x) = p^x (1 - p)^{1-x} \text{ for } x = 0, 1; \text{ and } 0 < p < 1$$

 Graph :



5. Binomial Distribution

Binomial Distribution

 **Definition:**

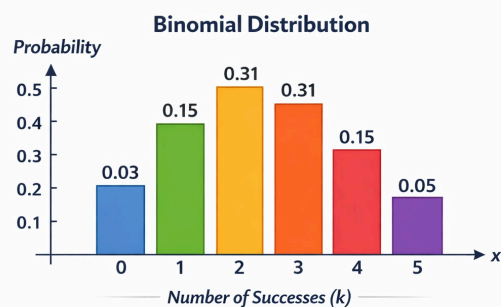
It counts **number of successes in fixed trials**

 **Example:**

- 10 coin toss → how many heads?
- 5 exams → how many passed?

 **Formula:**

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$
$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$



6. Log-Normal Distribution

Log-Normal Distribution

 **Meaning:**

If $\log(x)$ is normal $\rightarrow$ x is log-normal

 **Example:**

- income
- stock prices
- file sizes



Shape:

- Right-skewed (long tail)



Python Code :

 **Python**

```
import numpy as np
import matplotlib.pyplot as plt

data = np.random.lognormal(0, 1, 1000)
plt.hist(data, bins=30)
plt.title("Log-Normal Distribution")
plt.show()
```



Formula:

Log-Normal Distribution

$$\text{Mean: } \mu_x = e^{\mu + \sigma^2/2}$$

$$\text{Variance: } \sigma_x^2 = (e^{\sigma^2} - 1)e^{2\mu + \sigma^2}$$

$$f(x) = \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x - \mu)^2}{2\sigma^2}}, x > 0$$

7. Power Law Distribution

Power Law Distribution



Meaning:

Few values are very large

- Most values are very small

Example:

- population of cities
 - followers on social media
 - earthquakes
-

Shape:

- Long tail

Formula:

Power law distribution

- Straight line on a log-log plot

$$\ln(p(x)) = c - \alpha \ln(x)$$

- Exponentiate both sides to get that $p(x)$, the probability of observing an item of size 'x' is given by

$$p(x) = Cx^{-\alpha}$$

normalization
constant (probabilities over
all x must sum to 1)

power law exponent α

 Graph:

The Long Tail Power-Law Distribution



8. Box-Cox Transform

Box-Cox Transform

 Purpose:

- Make data normal
- Reduce skewness

 Python Code :

</> Python

```
from scipy import stats
import numpy as np

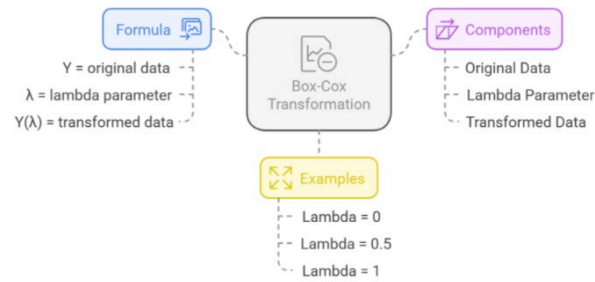
data = np.random.lognormal(0,1,100)

transformed, _ = stats.boxcox(data)

print(transformed[:5])
```

 **Graph:**

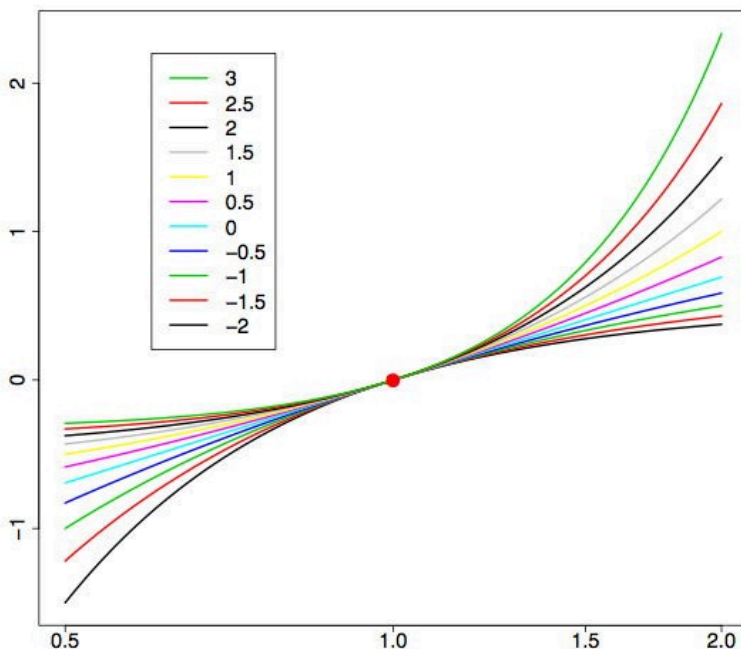
Understanding Box-Cox Transformation



 **Formula:**

$$\begin{cases} y = \frac{x^\lambda - 1}{\lambda} & \text{where } \lambda \neq 0 \\ y = \ln x & \text{where } \lambda = 0 \end{cases}$$

 **Diagram:**



9. Poisson Distribution

Poisson Distribution

Definition:

Used to count events in fixed time/space.

Example:

- calls per hour
- accidents per day
- messages per minute

Python Code :

Python

```
import numpy as np
import matplotlib.pyplot as plt

data = np.random.poisson(lam=5, size=1000)

plt.hist(data, bins=20)
plt.title("Poisson Distribution")
plt.show()
```

 Formula:

THE POISSON PROBABILITY FORMULA

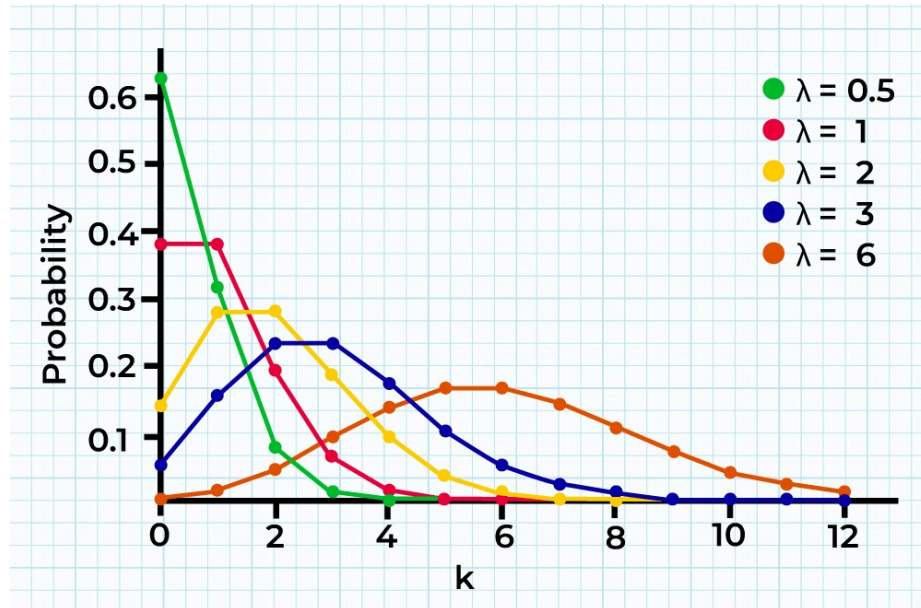
$$P(X = K) = \frac{e^{-\lambda} \lambda^K}{K!}$$

Where: λ = average number of events per interval

K = actual number of events

e = Euler's number ($\approx 2,718$)

 Graph:



10. Z-score Probability

Z-score

 **Formula:**

$$Z = (X - \text{mean}) / \text{std}$$

 **Meaning:**

- $Z = 0 \rightarrow$ average
- $Z > 0 \rightarrow$ above average
- $Z < 0 \rightarrow$ below average

 **Example:**

If score = 80, mean = 70, std = 5

$$Z = (80-70)/5 = 2$$

 Python Code :

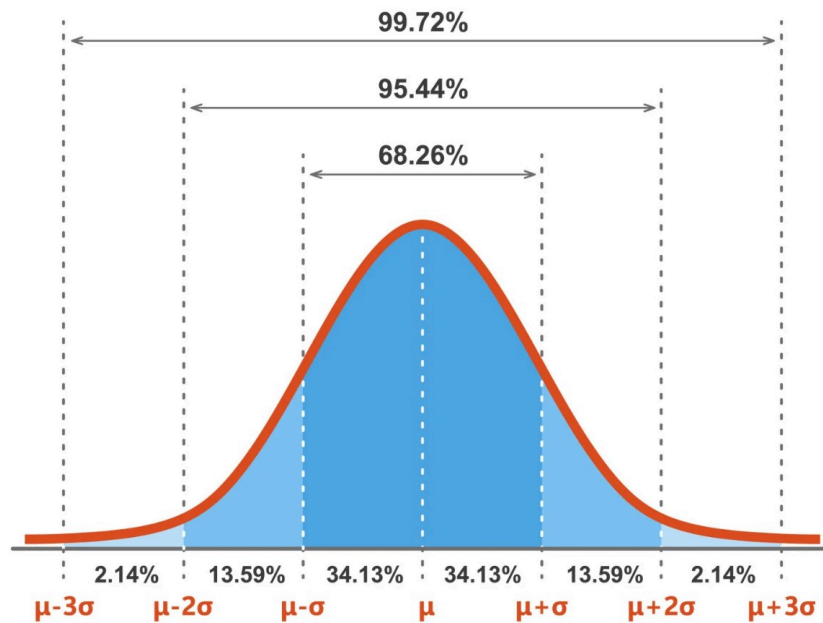
 **Python**

```
import numpy as np
from scipy import stats

data = [10,20,30,40,50]
z = stats.zscore(data)

print(z)
```

 Graph:



11. Difference Between Probability Density Function (PDF) and Cumulative Distribution Function (CDF)

Point	PDF (Probability Density Function)	CDF (Cumulative Distribution Function)
1. Full Form	Probability Density Function	Cumulative Distribution Function
2. Meaning	Shows probability density at different values.	Shows cumulative probability up to a value.
3. Purpose	Describes how probability is distributed.	Calculates total probability up to a point.
4. Formula	$(f(x))$	$(F(x)=P(X \leq x))$
5. Probability	Gives probability density, not direct probability.	Gives cumulative probability.
6. Output Range	Can be greater than 1 for some values.	Always between 0 and 1 .
7. Graph Shape	Bell-shaped or smooth curve.	Increasing S-shaped (or steadily increasing) curve.
8. Total Area	Total area under the curve is 1 .	Final value reaches 1 .

9. Nature	Does not decrease probability directly.	Always increases as (x) increases.
10. Used For	Finding probability over an interval.	Finding probability up to a given value.

 **Graph:**

